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Memory structure

Abstract

A memory structure includes a static random access memory (SRAM) cell having a first pass-gate transistor and a second pass-gate transistor, a word-line conductor extending in a first direction, a first source/drain contact, a second source/drain contact, a bit-line conductor in a second direction, and a bit-line-bar conductor extending in the second direction. The second direction is perpendicular to the first direction. The word-line conductor is over and electrically connected to gate electrodes of the first pass-gate transistor and the second pass-gate transistor. The first source/drain contact is under and electrically connected to a source/drain feature of the first pass-gate transistor. The second source/drain contact is under and electrically connected to a source/drain feature of the second pass-gate transistor. The bit-line conductor is under and electrically connected to the first source/drain contact. The bit-line conductor is under and electrically connected to the second source/drain contact.

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Background/Summary

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed.

(2) As integrated circuit (IC) technologies progress towards smaller technology nodes, gate-all-around (GAA) transistors have been incorporated into memory devices (including, for example,

static random-access memory, or SRAM, cells) to reduce chip footprint while maintaining reasonable processing margins.
(3) However, as GAA transistors and circuit cells continue to be scaled down, interconnection routing for memory array uses too many routing resources and therefore impact the cell scaling as well as cell performance. Accordingly, although existing technologies for fabricating memory array including GAA transistors have been generally adequate for their intended purposes, they have not been entirely satisfactory in all respects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1** is a fragmentary diagrammatic top view of an integrated circuit (IC) chip, in portion or entirety, in accordance with some embodiments of the present disclosure.
(3) FIG. **2** is a fragmentary diagrammatic top view of an array of SRAM cells that can be implemented in the memory region of FIG. **1**, in accordance with some alternative embodiments of the present disclosure.
(4) FIGS. **3** and **4** are circuit diagrams of an SRAM circuit that can be implemented in an SRAM cell of an array in the memory region of FIG. **1**, in accordance with some alternative embodiments of the present disclosure.
(5) FIG. **5** is a perspective view of a GAA transistor in an array of SRAM cells, in accordance with some embodiments of the present disclosure.
(6) FIG. **6** shows a cross sectional view of a memory structure for illustrating front-side interconnection and back-side interconnection, in accordance with some embodiments of the present disclosure.
(7) FIGS. **7**A and **7**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(8) FIG. **7**C illustrates a cross sectional view of the SRAM cell along a line C-C′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure.
(9) FIG. **7**D illustrates a cross sectional view of the SRAM cell along a line D-D′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure.
(10) FIG. **7**E illustrates a cross sectional view of the SRAM cell along a line E-E′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure.
(11) FIG. **7**F illustrates a cross sectional view of the SRAM cell along a line F-F′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure.
(12) FIG. **7**G illustrates a cross sectional view of the SRAM cell along a line G-G′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure.
(13) FIG. **8** illustrates a cross sectional view of tap structures in FIG. **2** for connecting the metal conductor at the back-side to a front-side metal conductor at the front-side, in accordance with some embodiments of the present disclosure.
(14) FIGS. **9**A and **9**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(15) FIGS. **10**A and **10**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments

of the present disclosure.
(16) FIG. **10**C illustrates a cross sectional view of the SRAM cell along a line C-C′ in FIGS. **10**A and **10**B, in accordance with some embodiments of the present disclosure.
(17) FIGS. **11**A and **11**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(18) FIG. **11**C illustrates a cross sectional view of the SRAM cell along a line C-C′ in FIGS. **11**A and **11**B, in accordance with some embodiments of the present disclosure.
(19) FIGS. **12**A and **12**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(20) FIG. **12**C illustrates a cross sectional view of the SRAM cell along a line C-C′ in FIGS. **12**A and **12**B, in accordance with some embodiments of the present disclosure.
(21) FIGS. **13**A and **13**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(22) FIGS. **14**A and **14**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(23) FIGS. **15**A and **15**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(24) FIGS. **16**A and **16**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(25) FIGS. **17**A and **17**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.
(26) FIGS. **18**A and **18**B illustrate top views (or layouts) of an SRAM cell that can be one embodiment of the SRAM cells implemented in the array, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(27) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(28) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(29) The present disclosure is generally related to memory structures, and more particularly to an array of static random-access memory (SRAM) cells having field-effect transistors (FETs), such as three-dimensional gate-all-around (GAA) transistors, in an integrated circuit (IC) structure. Generally, a GAA transistor may include a plurality of vertically stacked sheets (e.g., nanosheets), wires (e.g., nanowires), or rods (e.g., nanorods) in a channel region of the transistor, thereby allowing better gate control, lowered leakage current, and improved scaling capability for various IC applications.
(30) The gate-all-around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.
(31) Embodiments of the present disclosure offer advantages over the existing art, though it should be understood that other embodiments may offer different advantages, not all advantages are necessarily discussed herein, and no particular advantage is required for all embodiments. For example, embodiments discussed herein include an array of SRAM cells with bit-line conductors and bit-line-bar conductors under the SRAM cells (more specifically, functional transistors), such that improve cell performance and reduce routing complexity for SRAM cells. The details of the present disclosure are described below in conjunction with the accompanying drawings, which illustrate the layout and structure of circuit cells, according to some embodiments.
(32) FIG. **1** is a fragmentary diagrammatic top view of an integrated circuit (IC) chip **10**, in portion or entirety, in accordance with some embodiments of the present disclosure. IC chip **10** may include various passive microelectronic devices and active microelectronic devices, such as resistors, capacitors, inductors, diodes, p-type field effect transistors (PFETs), n-type field effect transistors (NFETs), metal-oxide semiconductor field effect transistors (MOSFETs), CMOS transistors, bipolar junction transistors (BJTs), laterally diffused MOS (LDMOS) transistors, high voltage transistors, high frequency transistors, other suitable components, or combinations thereof. The various microelectronic devices can be configured to provide IC chip **10** with functionally distinct regions, such as a core region (also referred to as a logic region), a memory region (e.g., a static random access memory (SRAM) region), an analog region, a peripheral region (also referred to as an input/output (I/O) region), a dummy region, and/or other suitable region. In some embodiments, IC chip **10** includes a memory region **20** and a logic region **30**. Memory region **20** can include an array of memory cells, each of which includes transistors and interconnection structures (also referred to as routing structures) that combine to provide a storage device and/or a storage function, such as a flip flop, a latch, other suitable memory devices, or combinations thereof. In some embodiments, memory region **20** is configured with static random-access memory (SRAM) cells, dynamic random-access memory (DRAM) cells, non-volatile random-access memory (NVRAM) cells, flash memory cells, other suitable memory cells, or combinations thereof. Logic region **30** can include an array of standard cells, each of which includes transistors and interconnection structures that combine to provide a logic device and/or a logic function, such as an inverter, an AND gate, an NAND gate, an OR gate, an NOR gate, a NOT gate, an XOR gate, an XNOR gate, other suitable logic devices, or combinations thereof. FIG. **1** has been simplified for the sake of clarity to better understand the inventive concepts of the present disclosure. Additional features can be added in IC chip **10**, and some of the features described herein can be replaced, modified, or eliminated in other embodiments of IC chip **10**.
(33) FIG. **2** is a fragmentary diagrammatic top view of an array **100** of SRAM cells **101** that can be

implemented in the memory region **20** of FIG. **1**, in accordance with some alternative embodiments of the present disclosure. FIG. **2** has been simplified for the sake of clarity to better understand the inventive concepts of the present disclosure. Additional features can be added in the array **100**, and some of the features described below can be replaced, modified, or eliminated in other embodiments of the array **100**.

(34) The array **100** includes SRAM cells **101** arranged with pluralities of columns and rows. Each of columns of the SRAM cells **101** includes a bit line pair extending in a Y-direction, such as a bit-line conductor (BL_**1**, BL_**2**, . . . , BL_N−1, BL_N) and a bit-line-bar conductor (also referred to as a complementary bit line) (BLB_**1**, BLB_**2**, BLB_N−1, BLB_N), that facilitate reading data from and/or writing data to respective SRAM cells **101** in true form and complementary form on a column-by-column basis. Each of rows of the SRAM cells **101** includes a word-line conductor (WL_**1**, WL_**2**, . . . , WL_M−1, WL_M) extending in an X-direction perpendicular to the Y-direction, that facilitates access to respective SRAM cells **101** on a row-by-row basis. Each of SRAM cells **101** is electrically connected to a respective bit-line conductor, a respective bit-line-bar conductor, and a respective word-line conductor, in which the bit-line conductors and the bit-line-bar conductors are electrically connected to a controller **102** and the word-line conductors are electrically connected to a controller **103**.

(35) The controllers **102** and **103** include any circuitry suitable to facilitate read/write operations from/to the SRAM cells **101**, including but not limited to, a column decoder circuit, a row decoder circuit, a column selection circuit, a row selection circuit, a read/write circuit (for example, configured to read data from and/or write data to the SRAM cells **101** corresponding to a selected bit line pair (in other words, a selected column)), other suitable circuit, or combinations thereof. In some implementations, the controllers **102** and **103** includes at least one sense amplifier configured to detect and/or amplify a voltage differential of a selected bit line pair. In some implementations, the sense amplifier is configured to latch or otherwise store data values of the voltage differential.

(36) The array **100** further includes edge cell regions **105**A and **105**B on edges of the array **100** in the Y-direction. The edge cell regions **105**A and **105**B include dummy cells for ensuring uniformity in performance of SRAM cells **101**. Dummy cells are configured physically and/or structurally similar to SRAM cells **106**, but do not store data. For example, dummy cells may include p-type wells, n-type wells, nanostructures, gate structures, source/drain features, and/or contact features.

(37) The array **100** also includes edge strap regions **104**A and **104**B on edges of the array **100** in the X-direction. The edge strap regions **104**A and **104**B does not contain SRAM cells and is used for implementing well pick-up structures or well strap cells configured to electrically couple a voltage to an n-well or a p-well of the SRAM cells **101**.

(38) In the present disclosure, some conductors for interconnection of the array **100** of the SRAM cells **101** are disposed under the SRAM cells **101** (on back-side of the SRAM cells **101**). For example, the bit-line conductors (BL_**1**, BL_**2**, . . . , BL_N−1, BL_N) and the bit-line-bar conductors (BLB_**1**, BLB_**2**, . . . , BLB_N−1, BLB_N) shown in FIG. **2** are disposed under and electrically connected to the SRAM cells **101**, will discussed in below. In order to electrically connect the bit-line conductors and the bit-line-bar conductors to the controller **102**, the bit-line conductors and the bit-line-bar conductors under the SRAM cells **101** are routed to front-side conductors over the SRAM cells **101** through tap structures (e.g., tap structures **2000**) located in the edge strap regions **104**A or **104**B, and then the front-side conductors are electrically connected to the controller **102**. In some embodiments, the conductors for word-lines, VDD lines, or VSS lines disposed under the SRAM cells **101** may also be routed to the front-side conductors over the SRAM cells **101** through the tap structures located in the edge strap regions **104**A/**104**B and the edge cell regions **105**A/**105**B. The details of the tap structures are described below.

(39) FIG. **3** and FIG. **4** are circuit diagrams of an SRAM circuit that can be implemented in an SRAM cell (e.g., the SRAM cell **101** in FIG. **2**) of an array in the memory region **20** of FIG. **1**, in accordance with some alternative embodiments of the present disclosure. The circuit diagram of

SRAM cell is merely exemplary, and in some embodiments, each of SRAM cells (e.g., SRAM cells **101**A to **101**K in FIGS. **7**A to **18**B) in the array is configured with an SRAM circuit similar to the SRAM cell **101** and as shown in FIG. **3** and FIG. **4**. For example, each of SRAM cells has a storage portion that includes a cross-coupled pair of inverters (also referred to as a latch), such as an Inverter-**1** and an Inverter-**2**. Inverter-**1** includes pull-up transistor PU-**1** and pull-down transistor PD-**1**, and Inverter-**2** includes pull-up transistor PU-**2** and pull-down transistor PD-**2**. Pass-gate transistor PG-**1** is connected to an output of Inverter-**1** and an input of Inverter-**2**, and pass-gate transistor PG-**2** is connected to an output of Inverter-**2** and an input of Inverter-**1**. In operation, pass-gate transistor PG-**1** and pass-gate transistor PG-**2** provide access to the storage portion of their respective SRAM cell (i.e., Inverter-**1** and Inverter-**2**) and can also be referred to as access transistors of their respective SRAM cell. Each of SRAM cells (e.g., the SRAM cells **101**A to **101**K in FIGS. **7**A to **18**B) is connected to and powered through a first power supply voltage, such as a positive power supply voltage, and a second power supply voltage, such as a ground voltage or a reference voltage (which can be an electrical ground). A gate of pull-up transistor PU-**1** interposes a source, which is electrically coupled to the first power supply voltage via a voltage node (or voltage source) VDD, and a first common drain (CD**1**) (i.e., a drain of pull-up transistor PU-**1** and a drain of pull-down transistor PD-**1**). A gate of pull-down transistor PD-**1** interposes a source, which is electrically coupled to the second power supply voltage via a voltage node (or voltage source) VSS, and the first common drain. A gate of pull-up transistor PU-**2** interposes a source, which is electrically coupled to the first power supply voltage via voltage node VDD, and a second common drain (CD-**2**) (i.e., a drain of pull-up transistor PU-**2** and a drain of pull-down transistor PD-**2**). A gate of pull-down transistor PD-**2** interposes a source, which is electrically coupled to the second power supply voltage via voltage node VSS, and the second common drain. The first common drain provides a storage node SN that stores data in true form, and the second common drain provides a storage node SNB that stores data in complementary form, or vice versa, in some embodiments. The gate of pull-up transistor PU**1** and the gate of pull-down transistor PD-**1** are coupled together and to the second common drain SD**2**, and the gate of pull-up transistor PU-**2** and the gate of pull-down transistor PD-**2** are coupled together and to the first common drain SD**1**. A gate of pass-gate transistor PG-**1** interposes a drain connected to a bit line node BLN, which is electrically coupled to a bit line BL, and a source, which is electrically coupled to the first common drain SD**1**. A gate of pass-gate transistor PG-**2** interposes a drain connected to a complementary bit line node BLBN, which is electrically coupled to a complementary bit line BLB, and a source, which is electrically coupled to the second common drain SD**2**. Gates of pass-gate transistors PG-**1**, PG-**2** are connected to and controlled by a word line WL, which allows selection of a respective SRAM cell, such as the SRAM cell **101**A, for reading and/or writing. In some embodiments, pass-gate transistors PG-**1**, PG-**2** provide access to storage nodes SN, SNB, respectively, each of which can store a bit (e.g., a logical 0 or a logical 1), during read operations and/or write operations. For example, pass-gate transistors PG-**1**, PG-**2** couple storage nodes SN, SNB, respectively, to bit line BL and bit line bar BLB in response to voltage applied to gates of pass-gate transistors PG-**1**, PG-**2** by word line WL. In some embodiments, SRAM cells are single-port SRAMs. In some embodiments, SRAM cells are configured as multi-port SRAMs, such as dual-port SRAMs, and/or with more or less transistors than depicted, such as 8T SRAMs. FIG. **3** and FIG. **4** have been simplified for the sake of clarity to better understand the inventive concepts of the present disclosure. Additional features can be added in the SRAM circuits of FIG. **3** and FIG. **4**, and some of the features described herein can be replaced, modified, or eliminated in other embodiments of the SRAM circuits of FIG. **3** and FIG. **4**.

(40) Each of the SRAM cells discussed above is constructed by transistors. The transistors may be planar transistors, fin field-effect transistor (FinFET) transistors, gate-all-around (GAA) transistors, nano-wire transistors, nano-sheet transistors, or a combination thereof. For the sake of providing an example, an exemplary GAA transistor is illustrated in FIG. **5**. However, it should be understood

that the application should not be limited to a particular type of device, except as specifically claimed.

(41) Referring to FIG. **5**, a perspective view of an exemplary GAA transistor **200** is illustrated. The GAA transistor **200** is formed over a substrate **202**. The substrate **202** may contains a semiconductor material, such as bulk silicon (Si). In some embodiments, after the resultant GAA transistor **200** is formed, the substrate **202** may be removed by a suitable process (e.g., a chemical mechanical polishing (CMP) process) for forming back-side interconnection.

(42) The GAA transistor **200** also includes one or more nanostructures **204** (dash lines) extending in an X-direction and vertically stacked (or arranged) in a Z-direction. More specifically, the nanostructures **204** are spaced from each other in the Z-direction. In some embodiments, the nanostructures **204** may also be referred to as channels, channel layers, nanosheets, or nanowires.

(43) The GAA transistor **200** further includes a gate structure **206** including a gate dielectric layer **208** and a gate electrode **210**. The gate dielectric layer **208** wraps around the nanostructures **204** and the gate electrode **210** wraps around the gate dielectric layer **208** (not shown in FIG. **5**, may refer to FIGS. **7**D, **7**F, and **7**G). As shown in FIG. **5**, gate spacers **212** are on sidewalls of the gate structure **206** and over the nanostructures **204** (not shown in FIG. **2**, may refer to FIGS. **7**F and **7**G). A gate top dielectric layer **214** is over the gate dielectric layer **208**, the gate electrode **210**, and the nanostructures **204**. The gate top dielectric layer **214** is used for contact etch protection layer.

(44) The GAA transistor **200** further includes source/drain features **216**. As shown in FIG. **5**, two source/drain features **216** are on opposite sides of the gate structure **206**. The nanostructures **204** (dash lines) extends in the X-direction to connect one source/drain feature **216** to the other source/drain feature **216**. The source/drain features **216** may also be referred to as source/drain, or source/drain regions. In some embodiments, source/drain feature(s) may refer to a source or a drain, individually or collectively dependent upon the context.

(45) Isolation feature **218** is over the substrate **202** and under the gate dielectric layer **208**, the gate electrode **210**, and the gate spacers **212**. The isolation feature **218** is used for isolating the GAA transistor **200** from other devices. In some embodiments, the isolation feature **218** may include different structures, such as shallow trench isolation (STI) structure, deep trench isolation (DTI) structure. Therefore, the isolation feature **218** is also referred as to as a STI feature or DTI feature.

(46) Generally, interconnection of devices and circuit cells are disposed over or at front-side of transistors to form desired circuit routing. As transistors and circuit cells continue to be scaled down, space for interconnection routing is also decreased. In order to achieve desired circuit routing, metal conductor width and conductor-to-conductor space are decreased, thereby increasing resistance and parasitic capacitance to impact performance of devices and circuit cells. In some embodiments of present disclosure, a part of interconnection of devices and circuit cells is disposed under or at back-side of transistors to improve upon the above issue. FIG. **6** shows a cross sectional view of a memory structure **300** for illustrating front-side interconnection and back-side interconnection, in accordance with some embodiments of the present disclosure. FIG. **6** also illustrates X-cut cross sectional view **300**A and Y-cut cross sectional view **300**B of the memory structure **300**. The memory structure **300** has device region **302** (also referred to as a device layer), back-side interconnection structure **304**, and front-side interconnection structure **306**. The device region **302** is the region where the transistors and main features of SRAM cells (e.g., the SRAM cells **101**A to **101**K in FIGS. **7**A to **18**B) are located, such as gate structures, nanostructures, source/drain features, and contact features. The device region **302** has front-side **302-1** and back-side **302-2**. The back-side interconnection structure **304** is under the device region **302** or at the back-side **302-2** of the device region **302**, and the front-side interconnection structure **306** is over the device region **302** or at the front side **302-1** of the device region **302**. The back-side interconnection structure **304** includes inter-metal dielectric (IMD) **308**, a via B_V**0**, and metal conductors B_M**1**. The front-side interconnection structure **306** includes IMD **310**, vias VG, V**0**, and V**1**, and metal conductors M**1** and M**2**. The vias and metal conductors in the IMD **308** and **310**

electrically couples various transistors and/or components (for example, gate structures, source/drain features, resistors, capacitors, and/or inductors) in the device region **302**, such that the various devices and/or components can operate as specified by design requirements of circuit cells (e.g., logic cells and memory cells). It should be noted that there may be more vias and metal conductors in the IMD **308** and **310** for connections. The IMD **308** and **310** may be multilayer structure, such as one or more dielectric layers.

(47) Since the back-side interconnection structure **304** is at the back-side **302-2** of the device region **302**, the IMD **308**, the via B_V**0**, and the metal conductors B_M**1** may also be referred to as back-side IMD, back-side via, and back-side metal conductors, respectively. Since the front-side interconnection structure **306** is at the front-side **302-1** of the device region **302**, the IMD **310**, the vias VG, V**0**, and V**1**, and the metal conductors M**1** and M**2** may also be referred to as front-side IMD, front-side vias, and front-side metal conductors, respectively. In some embodiments, the via VG are connected to the gate structures (gate electrodes) of the transistors. Therefore, the via VG are also referred to as gate vias or front-side gate via.

(48) The formation of the back-side interconnection structure **304** may include removing the substrate (if present) by CMP process, forming a back-side dielectric layer (not shown) under the device region **302** (or the back-side **302-2** of the device region **302**), forming back-side contacts (not shown) connected to the source/drain features in the device region **302** in the back-side dielectric layer, forming a first dielectric layer of the IMD **308** under the back-side dielectric layer, forming back-side first level vias (e.g., the via B_V**0**) in the first dielectric layer, forming a second dielectric layer of the IMD **308** under the first dielectric layer, forming back-side first level metal conductors (e.g., the metal conductors B_M**1**) in the second dielectric layer, and forming protection layer (may be multiple layers and include dielectric layers, poly layers, or combination) under the fourth dielectric layer. The formation of the front-side interconnection structure **306** is similar to that of back-side interconnection structure **304**, in which the difference is that the formation processes of the front-side interconnection structure **306** are performed at the front-side **302-1** of the device region **302**, and may not be described in detail herein.

(49) FIGS. **7**A and **7**B illustrate top views (or layouts) of an SRAM cell **101**A that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **7**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **7**B illustrates the features in the device region and the back-side interconnection structure.

(50) FIG. **7**C illustrates a cross sectional view of the SRAM cell **101**A along a line C-C′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure. FIG. **7**D illustrates a cross sectional view of the SRAM cell **101**A along a line D-D′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure. FIG. **7**E illustrates a cross sectional view of the SRAM cell **101**A along a line E-E′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure. FIG. **7**F illustrates a cross sectional view of the SRAM cell **101**A along a line F-F′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure. FIG. **7**G illustrates a cross sectional view of the SRAM cell **101**A along a line G-G′ in FIGS. **7**A and **7**B, in accordance with some embodiments of the present disclosure.

(51) As shown in FIGS. **7**A and **7**B, the SRAM cell **101**A has a cell boundary CB indicated by the dotted rectangular box and constructed by two cell long boundaries in the X-direction and two cell short boundaries in the Y-direction. Such SRAM cells **101**A are arranged in rows along the X-direction and in columns along the Y-direction. In that regard, the length of the cell long boundaries is also the pitch of the array of SRAM cells **101**A along the X-direction, and the length of the cell short boundaries is also the pitch of the array of SRAM cells **101**A along the Y-direction.

(52) The SRAM cell **101**A includes active areas, such as active areas **402-1** to **402-4**, (may be collectively referred to as the active areas **402**) that extend lengthwise in the Y-direction. Each of

active areas **402** includes channel regions, source regions, and drain regions (where source regions and drain regions are collectively referred to as source/drain regions herein) of transistors. The SRAM cell **101**A further includes gate structures, such as gate structures **404-1** to **404-4** (may be collectively referred to as the gate structures **404**) that extend lengthwise in the X-direction perpendicular to the Y-direction. The gate structures **404-1** to **404-4** are disposed over the channel regions of the respective active areas **402-1** to **402-4** (i.e., (vertically stacked) nanostructures **410**) and disposed between respective source/drain regions of the active areas **402-1** to **402-4** (i.e., source/drain features **412**N and **412**P). In some embodiments, the gate structures **404-1** to **404-4** wrap and/or surround suspended, vertically stacked nanostructures **410** in the channel regions of the active areas **402-1** to **402-4**, respectively (as shown in FIGS. **7**D, **7**F, and **7**G).

(53) The gate structure **404-1** extends across the active area **402-1** in the top view and engages the active area **402-1** to form the pass-gate transistor PG-**1**; the gate structure **404-2** extends across the active areas **402-1** and **402-2** in the top view and engages the active area **402-1** and **402-2** to form the pull-down transistor PD-**1** and the pull-up transistor PU-**1** respectively; the gate structure **404-3** extends across the active areas **402-3** and **402-4** in the top view and engages the active area **402-3** and **402-4** to form the pull-up transistor PU-**2** and the pull-down transistor PD-**2** respectively; and the gate structure **404-4** extends across the active area **402-4** in the top view and engages the active area **402-4** to form the pass-gate transistor PG-**2**. Further, the pull-down transistor PD-**1** and the pull-up transistor PU-**1** share the gate structure **404-2**, and the pull-down transistor PD-**2** and the pull-up transistor PU-**2** share the gate structure **404-3**, so that the gate structure **404-2** and the gate structure **404-3** are also referred to as common gates or shared gate structures.

(54) Similar to the isolation feature **218** discussed above, the SRAM cell **101**A further includes an isolation feature (or isolation structure) **414**. The isolation feature **414** may include silicon oxide, silicon nitride, silicon oxynitride, other suitable isolation material (for example, including silicon, oxygen, nitrogen, carbon, or other suitable isolation constituent), or combinations thereof. The isolation feature **414** may include different structures, such as shallow trench isolation (STI) structures, deep trench isolation (DTI) structures, and/or local oxidation of silicon (LOCOS) structures. In some embodiments, STI features include a multi-layer structure that fills the trenches, such as a silicon nitride comprising layer disposed over a thermal oxide comprising liner layer. In another example, STI features include a dielectric layer disposed over a doped liner layer (including, for example, boron silicate glass (BSG) or phosphosilicate glass (PSG)). In yet another example, STI features include a bulk dielectric layer disposed over a liner dielectric layer, where the bulk dielectric layer and the liner dielectric layer include materials depending on design requirements.

(55) Each of the transistors in the SRAM cell **101**A (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**) includes nanostructures **410** similar to the nanostructures **204** discussed above. As shown in FIGS. **7**D, **7**F, and **7**G, the nanostructures **410** are suspended. In some embodiments, three nanostructures **410** are vertically stacked (or vertically arranged) from each other in the Z-direction for one transistor. However, there may be another appropriate number of nanostructures in one transistor. For example, there may be from 2 to 6 nanostructures **410** in one transistor. The nanostructures **410** further extend lengthwise in the Y-direction (FIGS. **7**A, **7**F, and **7**G) and widthwise in the X-direction (FIGS. **7**A and **7**D). In some embodiments, each of the nanostructures **410** has a width W in the X-direction and in a range from about 4 nm to about 70 nm, as shown in FIG. **7**D. In some embodiments, each of the nanostructures **410** has a thickness T in the Z-direction and in a range from about 4 nm to about 8 nm, as shown in FIG. **7**D. As shown in FIG. **7**D, in each of the transistors in the SRAM cell **101**A, three nanostructures **410** are spaced from each other in the Z-direction by a distance S in a range from about 6 nm to about 15 nm. In some embodiments, the nanostructures **410** have vertically a pitch P in the Z-direction and in a range from about 10 nm to about 23 nm. The nanostructures **410** may include a semiconductor material, such as silicon,

germanium, silicon carbide, silicon phosphide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide, silicon germanium (SiGe), SiPC, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP. In some embodiments, the nanostructures **410** include silicon for n-type transistors. In other embodiments, the nanostructures **410** include silicon germanium for p-type transistors. In some embodiments, the nanostructures **410** are all made of silicon, and the type of the transistors depend on work function metal layer wrapping around the nanostructures **410**. In some embodiments, the nanostructures **410** are epitaxially grown using a deposition technique such as epitaxial growth, vapor-phase epitaxy (VPE), molecular beam epitaxy (MBE), although other deposition processes, such as chemical vapor deposition (CVD), low pressure CVD (LPCVD), atomic layer deposition (ALD), ultrahigh vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), a combination thereof, or the like, may also be utilized.
(56) In some embodiments, each of the gate structures **404-1** to **404-4** has a gate length in the Y-direction and in a range from about 6 nm to about 20 nm. Each of the gate structures **404-1** to **404-4** has a gate dielectric layer **406** and a gate electrode layer **408**. The gate dielectric layers **406** wrap around each of the nanostructures **410** and the gate electrodes layer **408** wrap around the gate dielectric layer **406**. In some embodiments, the gate structures **404** each further includes an interfacial layer (such as having silicon dioxide, silicon oxynitride, or other suitable materials) between the gate dielectric layer **406** and the nanostructures **410**. The gate dielectric layers **406** may include oxide with nitrogen doped dielectric material (initial layer) combined with metal content high-K dielectric material (K value (dielectric constant)>13). For example, gate dielectric layers **406** may include hafnium oxide (HfO_2), which has a dielectric constant in a range from about 18 to about 40. Alternatively, the gate dielectric layers **406** may include other high-K dielectrics, such as TiO_2, HfZrO, Ta_2O_3, $HfSiO_4$, ZrO_2, $ZrSiO_2$, LaO, AlO, ZrO, TiO, Ta_2O_5, Y_2O_3, $SrTiO_3$ (STO), $BaTiO_3$ (BTO), BaZrO, HfLaO, HfSiO, LaSiO, AlSiO, HfTaO, HfTiO, $(Ba,Sr)TiO_3$ (BST), Al_2O_3, Si_3N_4, oxynitrides (SiON), combinations thereof, or other suitable material. The gate dielectric layers **406** may be formed by chemical oxidation, thermal oxidation, ALD, CVD, and/or other suitable methods.
(57) The gate electrode layer **408** is formed to wrap around the gate dielectric layer **406** and the center portions of the nanostructures **410**, as shown in FIGS. **7**F and **7**G. In some embodiments, the gate electrode layer **408** may include an n-type work function metal layer for n-type transistor or a p-type work function metal layer for p-type transistor. In an embodiment the n-type work function metal layer is a material such as W, Cu, AlCu, TiAlC, TiAlN, TiAl, Pt, Ti, TiN, Ta, TaN, Co, Ni, Ag, Al, TaAl, TaAlC, TaC, TaCN, TaSiN, Mn, Zr, other suitable n-type work function materials, or combinations thereof. For example, the n-type work function metal layer may be deposited utilizing ALD, CVD, or the like. However, any suitable materials and processes may be utilized to form the n-type work function metal layer. In an embodiment, the p-type work function metal layer may be formed from a metallic material such as W, Al, Cu, TiN, Ti, TiAlN, TiAl, Pt, Ta, TaN, Co, Ni, TaC, TaCN, TaSiN, $TaSi_2$, $NiSi_2$, Mn, Zr, $ZrSi_2$, Ru, AlCu, Mo, $MoSi_2$, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like. Additionally, the p-type work function metal layer may be deposited using a deposition process such as ALD, CVD, or the like, although any suitable deposition process may be used.
(58) In some embodiments, the gate electrode layer **408** may include a single layer or alternatively a multi-layer structure. In some embodiments, the gate electrode layer **408** may further include a capping layer, a barrier layer, and a fill material (not shown). The capping layer may be formed adjacent to the gate dielectric layers **406** and may be formed from a metallic material such as TaN, Ti, TiAlN, TiAl, Pt, TaC, TaCN, TaSiN, Mn, Zr, TiN, Ru, Mo, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates,

oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like. The metallic material may be deposited using a deposition process such as ALD, CVD, or the like, although any suitable deposition process may be used. The barrier layer may be formed adjacent the capping layer, and may be formed of a material different from the capping layer. For example, the barrier layer may be formed of a material such as one or more layers of a metallic material such as TiN, TaN, Ti, TiAlN, TiAl, Pt, TaC, TaCN, TaSiN, Mn, Zr, Ru, Mo, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like. The barrier layer may be deposited using a deposition process such as atomic layer deposition, chemical vapor deposition, or the like, although any suitable deposition process may be used.
(59) The SRAM cell **101**A further includes gate top dielectric layers **416** are over the gate dielectric layers **406**, the gate electrodes **408**, and the nanostructures **410**. The gate top dielectric layers **416** are similar to the gate top dielectric layer **214** discussed above. The gate top dielectric layer **416** is used for contact etch protection layer.
(60) In some embodiments, the gate top dielectric layer **416** has a thickness in a range from about 2 nm to about 60 nm. The material of gate top dielectric layer **416** is selected from a group consisting of oxide, SiOC, SiON, SiOCN, nitride base dielectric, metal oxide dielectric, Hf oxide (HfO.sub.2), Ta oxide (Ta.sub.2O.sub.5), Ti oxide (TiO.sub.2), Zr oxide (ZrO.sub.2), Al oxide (A.sub.12O.sub.3), Y oxide (Y.sub.2O.sub.3), combinations thereof, or other suitable material.
(61) As shown in FIG. **7**D, gate end dielectrics **418** are at ends of the gate structures **404**. The gate end dielectrics **418** are used for separating the gate structures **404** aligned in the −direction. For example, the gate end dielectrics **320** separate the gate structures **404-1** and **404-3**, as shown in FIG. **7**D. The material of the gate end dielectrics **418** is selected from a group consisting of Si.sub.3N.sub.4, nitride-base dielectric, carbon-base dielectric, high K material (K>=9), or a combination thereof.
(62) The SRAM cell **101**A further includes gate spacers **420** are on sidewalls of the gate structures **404** and over the nanostructures **410**, as shown in FIGS. **7**F and **7**G. More specifically, the gate spacers **420** are over the nanostructures **410** and on top sidewalls of the gate structures **404**, and thus are also referred to as gate top spacers or top spacers. The gate spacers **420** may include multiple dielectric materials and be selected from a group consisting of silicon nitride (Si.sub.3N.sub.4), silicon oxide (SiO.sub.2), silicon carbide (SiC), silicon oxycarbide (SiOC), silicon oxynitride (SiON), silicon oxycarbon nitride (SiOCN), carbon doped oxide, nitrogen doped oxide, porous oxide, air gap, or a combination thereof. In some embodiments, the gate spacers **420** may include a single layer or a multi-layer structure.
(63) As shown in FIGS. **7**F and **7**G, the SRAM cell **101**A further includes inner spacers **422** on the sidewalls of the gate structures **404** and below the topmost nanostructures **410**. Furthermore, the inner spacers **422** are laterally between the source/drain features **412**N (or **412**P) and the gate structures **404**. The inner spacers **422** are also vertically between adjacent nanostructures **410**. The inner spacers **422** may include a dielectric material having higher K value (dielectric constant) than the gate spacers **420** and be selected from a group consisting of silicon nitride (Si.sub.3N.sub.4), silicon oxide (SiO.sub.2), silicon carbide (SiC), silicon oxycarbide (SiOC), silicon oxynitride (SiON), silicon oxycarbon nitride (SiOCN), air gap, or a combination thereof. In some embodiments, the gate spacers **420** and the inner spacers **422** have a thickness in the Y-direction and in a range from about 4 nm to about 12 nm. In some embodiments, the thickness of the gate spacers **420** in the Y-direction and the thickness of the inner spacers **422** in the Y-direction are the same. In other embodiments, the thickness of the gate spacers **420** in the Y-direction is less than the thickness of the inner spacers **422** in the Y-direction due to the gate spacers **420** are trimmed during sequent processes for forming source/drain contacts.
(64) Referring to FIGS. **7**C and **7**E to **7**G, the SRAM cell **101**A further includes source/drain

features **412**N and source/drain features **412**P in the source/drain regions of the active areas **402**. The source/drain features **412**N are disposed over both sides of the respective gate structure **404** and connected by the nanostructures **410** to form n-type transistor (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**). Similarly, the source/drain features **412**P are disposed over both sides of the respective gate structure **404** and connected by the nanostructures **410** to form p-type transistor (e.g., the pull-up transistors PU-**1** and PU-**2**).
(65) The source/drain features **412**N and **412**P may be formed by using epitaxial growth. In some embodiments, the source/drain features **412**N may include epitaxially-grown material selected from a group consisting of SiP, SiC, SiPC, SiAs, Si, or a combination thereof. In some embodiments, the epitaxially-grown material of the source/drain features **412**N may be doped with phosphorus (or arsenic, or both) having a doping concentration in a range from about 2×10.sup.19/cm.sup.3 to 3×10.sup.21/cm.sup.3. In some embodiments, the source/drain features **412**P may include epitaxially-grown material selected from a group consisting of boron-doped SiGe, boron-doped SiGeC, boron-doped Ge, boron-doped Si, boron and carbon doped SiGe, or a combination thereof. In some embodiments, the epitaxially-grown material of the source/drain features **412**P may be doped with boron having a doping concentration in a range from about 1×10.sup.19/cm.sup.3 to 6×10.sup.20/cm.sup.3.
(66) As shown in FIGS. **7**C and **7**E to **7**G, the SRAM cell **101**A further includes silicide features **424** over the source/drain features **412**N and **412**P. The silicide features **424** may include titanium silicide (TiSi), nickel silicide (NiSi), tungsten silicide (WSi), nickel-platinum silicide (NiPtSi), nickel-platinum-germanium silicide (NiPtGeSi), nickel-germanium silicide (NiGeSi), ytterbium silicide (YbSi), platinum silicide (PtSi), iridium silicide (IrSi), erbium silicide (ErSi), cobalt silicide (CoSi), or other suitable compounds.
(67) Referring to FIGS. **7**A to **7**C and **7**E to **7**G, the SRAM cell **101**A further includes source/drain contacts **430** (including source/drain contacts **430-1** to **430-6**) in an inter-layer dielectric (ILD) layer **426** and source/drain contacts **432** (including source/drain contacts **432-1** to **432-4**) in a dielectric layer **428**. As shown in FIG. **7**A, the source/drain contacts **430** and **432** extend lengthwise in the X-direction. The source/drain contacts **430** are self-aligned source/drain contacts. This means that the source/drain contacts **430** are formed by using the gate spacers **420** as mask. Therefore, the source/drain contacts **430** are in direct contact with the gate spacers **420**, as shown in FIGS. **7**F and **7**G. In some embodiments, the gate spacers **420** are trimmed due to the gate spacers **420** serving as the mask for forming the source/drain contacts **430**. Therefore, the thickness of the gate spacers **420** in the Y-direction is less than the thickness of the inner spacers **422** in the Y-direction, as discussed above.
(68) Furthermore, each of the source/drain contacts **430** is over and electrically connected to the respective source/drain features **412**N/**412**P and each of the source/drain contacts **432** is under electrically connected to the respective source/drain features **412**N/**412**P. Specifically, as shown in FIGS. **7**A, the source/drain contact **430-1** is over and electrically connected to the source/drain feature **412**P of the pull-up transistor PU-**2**; the source/drain contact **430-2** is over and electrically connected to the source/drain feature **412**N of the pull-down transistor PD-**2**; the source/drain contact **430-3** is over and electrically connected to the source/drain feature **412**N of the pass-gate transistor PG-**1** and pull-down transistor PD-**1** (also referred to as common source/drain or common drain) and the source/drain feature **412**P of the pull-up transistor PU-**1**; the source/drain contact **430-4** is over and electrically connected to the source/drain feature **412**N of the pass-gate transistor PG-**2** and pull-down transistor PD-**2** (also referred to as common source/drain or common drain) and the source/drain feature **412**P of the pull-up transistor PU-**2**; the source/drain contact **430-5** is over and electrically connected to the source/drain feature **412**N of the pull-down transistor PD-**1**; and the source/drain contact **430-6** is over and electrically connected to the source/drain feature **412**P of the pull-up transistor PU-**1**. As shown in FIG. **7**B, the source/drain contact **432-1** is under and electrically connected to the source/drain feature **412**N of the pass-gate

transistor PG-**1**; the source/drain contact **432-2** is under and electrically connected to the source/drain feature **412**P of the pull-up transistor PU-**2**; the source/drain contact **432-3** is under and electrically connected to the source/drain feature **412**P of the pull-up transistor PU-**1**; and the source/drain contact **432-4** is under and electrically connected to the source/drain feature **412**N of the pass-gate transistor PG-**2**. In some embodiments, the source/drain contacts **430** may be referred to as front-side source/drain contacts due to the source/drain contacts **430** are over the source/drain features **412**N/**412**P. In some embodiments, the source/drain contacts **432** may be referred to as back-side source/drain contacts due to the source/drain contacts **432** are under the source/drain features **412**N/**412**P.

(69) The SRAM cell **101**A further includes butted contacts **434-1** and **434-2**. As shown in FIG. **7**A, the butted contact **434-1** is over the source/drain contact **430-3** and the gate structure **404-3**, and the butted contact **434-2** is over the source/drain contact **430-4** and the gate structure **404-2**. In some embodiments, the butted contact **434-1** electrically connects the source/drain contact **430-3** to the gate structure **404-3** and the butted contact **434-2** electrically connects the source/drain contact **430-4** to the gate structure **404-2**. The butted contacts **434-1** and the source/drain contact **430-3** may correspond to the storage node SN shown in FIG. **2** and the butted contacts **434-2** and the source/drain contact **430-4** may correspond to the storage node SNB shown in FIG. **2**. The source/drain contacts **430** and **432** and butted contacts **434-1** and **434-2** may each include a conductive material such as Al, Cu, W, Co, Ti, Ta, Ru, Rh, Ir, Pt, TiN, TiAl, TiAlN, TaN, TaC, combinations of these, or the like, although any suitable material may be deposited using a deposition process such as sputtering, CVD, electroplating, electroless plating, or the like. In some embodiments, the source/drain contacts **430** and **432** and butted contacts **434-1** and **434-2** may each include single conductive material layer or multiple conductive layers.

(70) As discussed above, the front-side interconnection structure is over the device region or at the front-side of the device region. The SRAM cell **101**A further includes a front-side interconnection structure **502** including vias **504** (including vias **504-1** to **504-4**), metal conductors **506** (including metal conductors **506-1** to **506-5**), vias **508** (including vias **508-1** to **508-4**), metal conductors **510** (including metal conductors **510-1** and **510-2**), gate vias **512** (including gate vias **512-1** and **512-2**), an ILD layer **514**, and an IMD layer **516**, which are over (or at the front-side of) the transistors in the SRAM cell **101**A (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).

(71) The metal conductors **506** are in a (front-side) metal layer ML**1** in the IMD layer **516** and extend lengthwise in the Y-direction. The metal conductors **510** are in a (front-side) metal layer ML**2** in the IMD layer **516** and extend lengthwise in the X-direction. The metal layer ML**1** is over the SRAM cell **101**A and the metal layer ML**2** is over the metal layer ML**1**, and thus the metal conductors **506** are over the transistors of the SRAM cell **101**A and the metal conductors **510** are over the metal conductors **506**. Each of the vias **504** in the ILD layer **514** are vertically between and electrically connected to the respective source/drain contact **430** and the respective metal conductor **506**. Each of the gate vias **512** in the ILD layer **514** are vertically between and electrically connected to the respective gate structure **404** and the respective metal conductor **506**. Each of the vias **508** in the IMD layer **516** are vertically between and electrically connected to the respective metal conductor **506** and the respective metal conductor **510**. In some embodiments, the vias **504**, the vias **508**, and the gate vias **512** may have circular shape in the top view. In other embodiments, the vias **504**, the vias **508**, and the gate vias **512** may have a rectangular shape in the top view.

(72) The vias **504**, the metal conductors **506**, the vias **508**, the metal conductors **510**, and the gate vias **512** may be respectively similar to the via V**0**, the metal conductors M**1**, the vias V**1**, the metal conductors M**2**, and the via VG discussed above. The vias **504** and **508**, the gate vias **512**, the metal conductors **506** and **510**, the ILD layer **514**, and IMD layer **516** may also be referred to as front-side vias, front-side gate vias, front-side metal conductors, front-side ILD layer, and front-side IMD

layer, respectively.
(73) In some embodiments, the metal conductor **510-2** serves as the word-line that is electrically connected to a controller (e.g., the controller **103** discussed above) and electrically connected to the gate structures (more specifically, the gate electrodes) of the pass-gate transistors in the same row of the array of the SRAM cell. As shown in FIGS. **7**A to **7**E, in the SRAM cell **101**A, the metal conductor **510-2** is over the pass-gate transistor PG-**1** and PG-**2**. The metal conductor **510-2** is electrically connected to the gate structure **404-1** of the pass-gate transistor PG-**1** through the via **508-1**, the metal conductor **506-4**, and the gate via **512-1**, and is electrically connected to the gate structure **404-4** of the pass-gate transistor PG-**2** through the via **508-4**, the metal conductor **506-5**, and the gate via **512-2**. In some embodiments, the metal conductor **510-2** may be referred to as (front-side) word-line conductor. In some embodiments, the metal conductors **506-4** and **506-5** may be referred to as word-line landing pads.
(74) In some embodiments, the metal conductors **506-1** and **506-3** serve as the VSS lines that are electrically coupled to a voltage node or voltage source (not shown) (e.g., the voltage node (or voltage source) VSS discussed above) and electrically connected to the source/drain features of the pull-down transistors in the same column of the array of the SRAM cell. As shown in FIGS. **7**A to **7**E, in the SRAM cell **101**A, the metal conductor **506-1** and **506-3** are respectively over the pull-down transistor PD-**1** and PD-**2**. The metal conductor **506-1** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**1** through the via **504-3** and the source/drain contact **430-5**; and the metal conductor **506-3** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**2** through the via **504-2** and the source/drain contact **430-2**. In some embodiments, the metal conductors **506-1** and **506-3** may be referred to as (front-side) VSS conductors or (front-side) VSS lines.
(75) As discussed above, the metal conductors **506-1** and **506-3** are electrically coupled to the voltage node (or voltage source) VSS. The metal conductor **510-1** is electrically connected to the metal conductors **506-1** and **506-3** to serve as a power mesh line for connecting the voltage node VSS to the metal conductors **506-1** and **506-3**. As shown in FIGS. **7**A to **7**E, in the SRAM cell **101**A, the metal conductor **510-1** is electrically connected to the metal conductor **506-1** through the via **508-2** and electrically connected to the metal conductor **506-3** through the via **508-3**. In some embodiments, the metal conductor **510-1** may be referred to as a (front-side) VSS power mesh line or a VSS power mesh conductor.
(76) In some embodiments, the metal conductor **506-2** serves as the VDD line that is electrically coupled to a voltage node or voltage source (not shown) (e.g., the voltage node (or voltage source) VDD discussed above) and electrically connected to the source/drain features of the pull-up transistors in the same column of the array of the SRAM cell. As shown in FIGS. **7**A to **7**E, in the SRAM cell **101**A, the metal conductor **506-2** is over the pull-up transistor PU-**1** and PU-**2**. The metal conductor **506-2** is electrically connected to the source/drain features **412**P of the pull-up transistor PU-**1** through the via **504-4** and the source/drain contact **430-6**, and is electrically connected to the source/drain features **412**P of the pull-up transistor PU-**2** through the via **504-1** and the source/drain contact **430-1**. In some embodiments, the metal conductor **506-2** may be referred to as the (front-side) VDD conductors or the (front-side) VDD lines.
(77) As discussed above, the back-side interconnection structure is under the device region or at the back-side of the device region. The SRAM cell **101**A further includes a back-side interconnection structure **602** including vias **604** (including vias **604-1** to **604-4**), metal conductors **606** (including metal conductors **606-1** to **606-3**), and an IMD layer **608**, which are under (or at the back-side of) the transistors in the SRAM cell **101**A (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).
(78) The metal conductors **606** are in a (back-side) metal layer BML**1** in the IMD layer **608** and extend lengthwise in the Y-direction. The metal layer BML**1** is under the SRAM cell **101**A, and thus the metal conductors **606** are under the transistors of the SRAM cell **101**A. Each of the vias

604 in the IMD layer **608** are vertically between and electrically connected to the respective source/drain contact **432** and the respective metal conductor **606**. In some embodiments, the vias **604** may have circular shape in the top view. In other embodiments, the vias **604** may have a rectangular shape in the top view.

(79) The vias **604** and the metal conductors **606** may be respectively similar to the via B_V**0** and the metal conductors B_M**1** discussed above. The vias **604**, the metal conductors **606**, and IMD layer **608** may also be referred to as back-side vias, back-side metal conductors, and back-side IMD layer, respectively

(80) In some embodiments, the metal conductors **606-1** and **606-3** respectively serve as the bit-line and the bit-line-bar that are electrically connected to a controller (e.g., the controller **102** discussed above) and electrically connected to the source/drain features of the pass-gate transistors in the same column of the array of the SRAM cell. As shown in FIGS. **7**A to **7**E, in the SRAM cell **101**A, the metal conductor **606-1** and **606-3** are respectively under the pass-gate transistor PG-**1** and PG-**2**. The metal conductor **606-1** is electrically connected to the source/drain feature **412**N of the pass-gate transistor PG-**1** through the via **604-1** and the source/drain contact **432-1**, and the metal conductor **606-3** is electrically connected to the source/drain feature **412**N of the pass-gate transistor PG-**2** through the via **604-4** and the source/drain contact **432-4**. In some embodiments, the metal conductors **606-1** and **606-3** may be respectively referred to as (back-side) bit-line conductor and (back-side) bit-line-bar conductor.

(81) In some embodiments, the metal conductor **606-2** also serves as the VDD line that is electrically coupled to a voltage node or voltage source (not shown) (e.g., the voltage node (or voltage source) VDD discussed above) and electrically connected to the source/drain features of the pull-up transistors in the same column of the array of the SRAM cell. As shown in FIGS. **7**A to **7**E, in the SRAM cell **101**A, the metal conductor **606-2** is under the pull-up transistor PU-**1** and PU-**2**. The metal conductor **606-2** is electrically connected to the source/drain features **412**P of the pull-up transistor PU-**1** through the via **604-3** and the source/drain contact **432-3**, and is electrically connected to the source/drain features **412**P of the pull-up transistor PU-**2** through the via **604-2** and the source/drain contact **432-2**. In some embodiments, the metal conductor **606-2** may also be referred to as (back-side) VDD conductors or (back-side) VDD lines.

(82) The ILD layer **426**, the dielectric **428**, the ILD layer **514**, the IMD **516**, and the IMD **608** each may include one or more dielectric layers including dielectric materials, such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fluoride-doped silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), a low-k dielectric material, other suitable dielectric material, or a combination thereof.

(83) The materials of the vias **504**, the metal conductors **506**, the vias **508**, the metal conductors **510**, the gate vias **512**, the vias **604**, and the metal conductors **606** are selected from a group consisting of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), titanium aluminum nitride (TiAlN), tungsten nitride (WN), tungsten (W), cobalt (Co), molybdenum (Mo), ruthenium (Ru), platinum (Pt), aluminum (Al), copper (Cu), other conductive materials, or a combination thereof.

(84) As shown in FIGS. **7**A to **7**E, the metal conductors **606-1** and **606-3** (serving as the bit-line and the bit-line-bar) are disposed at the back-side of the SRAM cell **101**A, so that the crowded space at the front-side interconnection structure in existing technologies are relieved to reduce the routing complexity of the SRAM cells. Furthermore, the metal conductor **510-2** (serving as the word-line) and the metal conductors **606-1** and **606-3** (serving as the bit-line and the bit-line-bar) may be designed with wider width, so that reducing the circuit resistance. The conductor-to-conductor space between the metal conductors may also be increased to reduce parasitic capacitance. Furthermore, in a cross sectional view of the SRAM cell **101**A, such as FIGS. **7**F and **7**G, it should be noted that the source/drain contacts **432** are not surrounded by the gate structures

404 in the Y-direction. This means that the current from the back-side conductors, such as the metal conductors **606-1** and **606-3** (which serve as the bit-line and the bit-line-bar) is not affected by the gate structures **404** during operation of the SRAM cell **101**A when passing through the source/drain contacts **432**. As such, the reliability of the read/write operations of the SRAM cell **101**A are improved. Therefore, the performance of the SRAM cell **101**A are improved.

(85) As discussed above, referring back to FIG. **2**, the metal conductors disposed under the SRAM cells may also be routed to the front-side conductors over the SRAM cells through the tap structures **2000** located in the edge strap regions and the edge cell regions. The metal conductors **606-1** and **606-3** in the SRAM cell **101**A (serving as the bit-line and the bit-line-bar) may extend in the Y-direction to the edge strap regions of the array constructed by the SRAM cells **101**A, and then may be routed to the front-side conductors through the tap structures (e.g., the tap structures **2000**).

(86) FIG. **8** illustrates a cross sectional view of tap structures **2000** in FIG. **2** for connecting the metal conductor **606-1** or **606-3** at the back-side to a front-side metal conductor at the front-side, in accordance with some embodiments of the present disclosure. As shown in FIG. **8**, a tap structure **2000** may electrically connect the metal conductor **606-1** or **606-3** to a front-side bit-line conductor **2012** electrically connected to the controller (e.g., the controller **102**). The tap structure **1016** may include a via **2004**, a tap via **2006**, a contact feature **2008**, a via **2010**. The via **2004** is at the back-side of the device region discussed above, and the via **2010** and the contact feature **2008** are at the front-side of the device region. In some embodiments, the via **2004** and the vias **604** are formed at the same fabrication operation, the contact feature **2008** and the source/drain contacts **430** are formed at the same fabrication operation, and the via **2010** and the via **504** are formed at the same fabrication operation. In some embodiments, the tap via **2006** is formed after the formation of the gate vias **512**. Such tap structure **2000** for the metal conductor **606-1** or **606-3** served as bit-line or bit-line-bar may be referred to as bit-line tap structure or bit-line-bar tap structure. It should be noted that the tap structure **2000** may be applied for connecting any back-side metal conductor to a front-side metal conductor. It should be noted that the tap structure **2000** may have more vias and more metal conductors.

(87) FIGS. **9**A and **9**B illustrate top views (or layouts) of an SRAM cell **101**B that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **9**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **9**B illustrates the features in the device region and the back-side interconnection structure.

(88) The SRAM cell **101**B is similar to the SRAM cell **101**A discussed above, except that no back-side vias are formed. The vias **432** shown in the SRAM cell **101**A are not formed in the SRAM cell **101**B. As shown in FIGS. **9**A and **9**B, the metal conductors **606-1** to **606-3** are in contact with the source/drain contacts **432**. As such, the metal conductor **606-1** is electrically connected to the source/drain feature **412**N of the pass-gate transistor PG-**1** through the source/drain contact **432-1**; the metal conductor **606-2** is electrically connected to the source/drain features **412**P of the pull-up transistor PU-**1** through the source/drain contact **432-3**, and is electrically connected to the source/drain features **412**P of the pull-up transistor PU-**2** through the source/drain contact **432-2**; and the metal conductor **606-3** is electrically connected to the source/drain feature **412**N of the pass-gate transistor PG-**2** through the source/drain contact **432-4**. In this embodiment, the cost of forming the vias **432** can be saved and the resistance of the back-side interconnection structure **602** can be reduced.

(89) FIGS. **10**A and **10**B illustrate top views (or layouts) of an SRAM cell **101**C that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **10**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal

conductors), and FIG. **10**B illustrates the features in the device region and the back-side interconnection structure. FIG. **10**C illustrates a cross sectional view of the SRAM cell **101**C along a line C-C′ in FIGS. **10**A and **10**B, in accordance with some embodiments of the present disclosure.
(90) The SRAM cell **101**C is similar to the SRAM cell **101**B discussed above, except that no back-side VDD line is formed. The source/drain contact **432-2**, source/drain contact the **432-3**, and the metal conductor **606-2** shown in the SRAM cell **101**B are not formed in the SRAM cell **101**C. As such, as shown in FIGS. **10**A to **10**C, the single metal conductor **506-2** serves as the VDD line that is electrically connected to the source/drain features **412**P of the pull-up transistor PU-**1** through the via **504-4**, and is electrically connected to the source/drain features **412**P of the pull-up transistor PU-**2** through the via **504-1**. In this embodiment, the cost of forming source/drain contact **432-2**, source/drain contact the **432-3**, and the metal conductor **606-2** can be reduced and the metal conductors **606-1** and **606-3** may be designed with a wider width (than that of the SRAM cell **101**A) to reduce circuit resistance.
(91) FIGS. **11**A and **11**B illustrate top views (or layouts) of an SRAM cell **101**D that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **11**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **11**B illustrates the features in the device region and the back-side interconnection structure. FIG. **11**C illustrates a cross sectional view of the SRAM cell **101**D along a line C-C′ in FIGS. **11**A and **11**B, in accordance with some embodiments of the present disclosure.
(92) The SRAM cell **101**D is similar to the SRAM cell **101**A discussed above, except that the SRAM cell **101**D further includes a metal conductor **702** in back-side interconnection structure **602**. The metal conductors **702** are in a (back-side) metal layer BML**2** in the IMD layer **608** and extend lengthwise in the X-direction. The metal layer BML**2** is under the metal layer BML**1**, and thus the metal conductors **702** are under the metal conductors **606**.
(93) In some embodiments, the metal conductor **702** serves as the word-line that is electrically connected to the metal conductor **510-2** (serving as the word-line) for a double world-line connection. The metal conductor **510-2** and **702** may extend in the X-direction to the edge cell regions (e.g., the edge cell regions **105**A or **105**B discussed above) of the array constructed by the SRAM cells **101**D, and then be electrically connected with each other through the tap structures (e.g., the tap structures **2000** discussed above) located at the edge cell regions. In some embodiments, the metal conductor **702** may be referred to as (back-side) word-line conductor.
(94) FIGS. **12**A and **12**B illustrate top views (or layouts) of an SRAM cell **101**E that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **12**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **12**B illustrates the features in the device region and the back-side interconnection structure. FIG. **12**C illustrates a cross sectional view of the SRAM cell **101**E along a line C-C′ in FIGS. **12**A and **12**B, in accordance with some embodiments of the present disclosure.
(95) The SRAM cell **101**E is similar to the SRAM cell **101**A discussed above, except that the SRAM cell **101**E further includes a via **802**, a metal conductor **804**, a via **806**, a the metal conductors **808**, which are over (or at the front-side of) the transistors in the SRAM cell **101**E (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).
(96) The metal conductor **804** are in a metal layer ML**3** in the IMD layer **516** and extend lengthwise in the Y-direction. The metal conductors **808** are in a metal layer ML**4** in the IMD layer **516** and extend lengthwise in the X-direction. The metal layer ML**3** is over the metal layer ML**2** and the metal layer ML**4** is over the metal layer ML**3**, and thus the metal conductors **804** are over the metal conductors **510-2** and the metal conductors **808** are over the metal conductor **804**. The via **802** in the IMD layer **516** is vertically between and electrically connected to the metal conductor

510-2 and the metal conductor **804**. The via **806** in the IMD layer **516** is vertically between and electrically connected to the metal conductor **804** and the metal conductor **808**. In some embodiments, the vias **802** and **804** may have circular shape in the top view. In other embodiments, the vias **802** and **804** may have a rectangular shape in the top view.

(97) In some embodiments, the metal conductor **808** serves as the word-line that is electrically connected to the metal conductor **510-2** (serving as the word-line) for a double world-line connection. The metal conductor **808** is electrically connected to the metal conductor **510-2** through the via **806**, the metal conductor **804**, and the via **802**. In some embodiments, the metal conductor **808** may be referred to as (front-side) word-line conductor.

(98) FIGS. **13**A and **13**B illustrate top views (or layouts) of an SRAM cell **101**F that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **13**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **13**B illustrates the features in the device region and the back-side interconnection structure.

(99) The SRAM cell **101**F is similar to the SRAM cell **101**A discussed above, except that the metal conductor for word-line is formed in the metal layer BML**1**, the metal conductors for VSS lines are formed in the back-side interconnection structure **602**, and no metal conductor for VDD line in the front-side interconnection structure **502**. The source/drain contacts **430**, the vias **504**, the metal conductors **506**, the vias **508**, and the metal conductors **510** shown in the SRAM cell **101**A are not formed in the SRAM cell **101**F. The front-side interconnection structure **502** of the SRAM cell **101**F further includes a metal conductor **902**, which is over (or at the front-side of) the transistors in the SRAM cell **101**F (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).

(100) The metal conductor **902** is in the metal layer ML**1** in the IMD layer **516** and extend lengthwise in the X-direction. In some embodiments, the metal conductor **902** serves as the word-line that is electrically connected to a controller (e.g., the controller **103** discussed above) and electrically connected to the gate structures (more specifically, the gate electrodes) of the pass-gate transistors in the same row of the array of the SRAM cell. As shown in FIGS. **13**A to **13**B, in the SRAM cell **101**F, the metal conductor **902** is over the pass-gate transistor PG-**1** and PG-**2**. The metal conductor **902** is electrically connected to the gate structure **404-1** of the pass-gate transistor PG-**1** through the gate via **512-1**, and is electrically connected to the gate structure **404-4** of the pass-gate transistor PG-**2** through the gate via **512-2**. In some embodiments, the metal conductor **902** may be referred to as (front-side) word-line conductor. In this embodiment, the metal conductor **902** may be designed with wider width (than the metal conductor **510-2** of the SRAM cell **101**A) to reduce circuit resistance.

(101) The SRAM cell **101**F further includes source/drain contacts **1002** (including source/drain contacts **1002-1** and **1002-2**), and vias **1004** (including vias **1004-1** and **1004-2**) and metal conductors **1006** (including metal conductors **1006-1** and **1006-2**) in the back-side interconnection structure **602**, which are under (or at the back-side of) the transistors in the SRAM cell **101**F (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).

(102) As shown in FIG. **13**B, the source/drain contacts **1002** extend lengthwise in the X-direction. The source/drain contact **1002-1** is under and electrically connected to the source/drain feature **412**N of the pull-down transistor PD-**1**; and the source/drain contact **1002-2** is under and electrically connected to the source/drain feature **412**N of the pull-down transistor PD-**2**.

(103) The metal conductors **1006** are in the metal layer BML**1** in the IMD layer **608** and extend lengthwise in the Y-direction. The metal conductors **1006** overlap cell short boundaries of the cell boundary CB of the SRAM cell **101**F in the Y-direction, as shown in FIG. **13**B. In some embodiments, the metal conductors **1006-1** and **1006-2** serve as the VSS lines that are electrically

coupled to a voltage node or voltage source (not shown) (e.g., the voltage node (or voltage source) VSS discussed above) and electrically connected to the source/drain features of the pull-down transistors in the same column of the array of the SRAM cell. As shown in FIGS. **13**A and **13**B, in the SRAM cell **101**F, the metal conductor **1006-1** and **1006-2** are respectively under the pull-down transistor PD-**1** and PD-**2**. The metal conductor **1006-1** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**1** through the via **1004-1** and the source/drain contact **1002-1**; and the metal conductor **1006-2** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**2** through the via **1004-2** and the source/drain contact **1002-2**. In some embodiments, the metal conductors **1006-1** and **1006-2** may be referred to as (back-side) VSS conductors or (back-side) VSS lines.

(104) FIGS. **14**A and **14**B illustrate top views (or layouts) of an SRAM cell **101**G that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **14**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **14**B illustrates the features in the device region and the back-side interconnection structure.

(105) The SRAM cell **101**G is similar to the SRAM cell **101**A discussed above, except that the metal conductors for VSS lines are formed in the back-side interconnection structure **602**, and no metal conductor for VDD line in the front-side interconnection structure **502**. The source/drain contacts **430**, the vias **504**, the metal conductors **506-1** to **506-3**, the vias **508-2** and **508-3**, and the metal conductor **510-1** shown in the SRAM cell **101**A are not formed in the SRAM cell **101**G. The SRAM cell **101**G further includes source/drain contacts **1102** (including source/drain contacts **1102-1** and **1102-2**), and vias **1104** (including vias **1104-1** and **1104-2**) and metal conductors **1106** (including metal conductors **1106-1** and **1106-2**) in the back-side interconnection structure **602**, which are under (or at the back-side of) the transistors in the SRAM cell **101**G (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).

(106) As shown in FIG. **14**B, the source/drain contacts **1102** extend lengthwise in the X-direction. The source/drain contact **1102-1** is under and electrically connected to the source/drain feature **412**N of the pull-down transistor PD-**1**; and the source/drain contact **1102-2** is under and electrically connected to the source/drain feature **412**N of the pull-down transistor PD-**2**.

(107) The metal conductors **1106** are in the metal layer BML**1** in the IMD layer **608** and extend lengthwise in the Y-direction. The metal conductors **1106** overlap cell short boundaries of the cell boundary CB of the SRAM cell **101**G in the Y-direction, as shown in FIG. **14**B. In some embodiments, the metal conductors **1106-1** and **1106-2** serve as the VSS lines that are electrically coupled to a voltage node or voltage source (not shown) (e.g., the voltage node (or voltage source) VSS discussed above) and electrically connected to the source/drain features of the pull-down transistors in the same column of the array of the SRAM cell. As shown in FIGS. **14**A and **14**B, in the SRAM cell **101**G, the metal conductor **1106-1** and **1106-2** are respectively under the pull-down transistor PD-**1** and PD-**2**. The metal conductor **1106-1** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**1** through the via **1104-1** and the source/drain contact **1102-1**; and the metal conductor **1106-2** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**2** through the via **1104-2** and the source/drain contact **1102-2**. In some embodiments, the metal conductors **1106-1** and **1106-2** may be referred to as (back-side) VSS conductors or (back-side) VSS lines.

(108) FIGS. **15**A and **15**B illustrate top views (or layouts) of an SRAM cell **101**H that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **15**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **15**B illustrates the features in the device region and the back-side

interconnection structure.
(109) The SRAM cell **101**H is similar to the SRAM cell **101**G discussed above, except that the SRAM cell **101**H further includes vias **1202** (including vias **1202-1** and **1202-2**) and a metal conductor **1204**, which are under (or at the back-side of) the transistors in the SRAM cell **101**H (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).
(110) The metal conductor **1204** is in a metal layer BML**2** in the IMD layer **608** and extend lengthwise in the X-direction. The metal layer BML**2** is under the metal layer BML**1**, and thus the metal conductor **1204** is under the metal conductors **606** and **1106**. The metal conductor **1204** is electrically connected to the metal conductors **1106-1** and **1106-2** to serve as the power mesh line for connecting the voltage node VSS to the metal conductors **1106-1** and **1106-3**. As shown in FIGS. **15**A and **15**B, in the SRAM cell **101**H, the metal conductor **1204** is electrically connected to the metal conductor **1106-1** through the via **1202-1** and electrically connected to the metal conductor **1106-2** through the via **1202-2**. In some embodiments, the metal conductor **1204** may be referred to as (front-side) VSS power mesh line or VSS power mesh conductor.
(111) FIGS. **16**A and **16**B illustrate top views (or layouts) of an SRAM cell **101**I that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **16**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **16**B illustrates the features in the device region and the back-side interconnection structure.
(112) The SRAM cell **101**I is similar to the SRAM cell **101**A discussed above, except that the metal conductors for VSS lines are formed in the back-side interconnection structure **602**. The vias **508-2** and **508-3**, and the metal conductor **510-1** shown in the SRAM cell **101**A are not formed in the SRAM cell **101**I. The SRAM cell **101**I further includes source/drain contacts **1302** (including source/drain contacts **1302-1** and **1302-2**), and vias **1304** (including vias **1304-1** and **1304-2**) and metal conductors **1306** (including metal conductors **1306-1** and **1306-2**) in the back-side interconnection structure **602**, which are under (or at the back-side of) the transistors in the SRAM cell **101**I (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).
(113) As shown in FIG. **16**B, the source/drain contacts **1302** extend lengthwise in the X-direction. The source/drain contact **1302-1** is under and electrically connected to the source/drain feature **412**N of the pull-down transistor PD-**1**; and the source/drain contact **1302-2** is under and electrically connected to the source/drain feature **412**N of the pull-down transistor PD-**2**.
(114) The metal conductors **1306** are in the metal layer BML**1** in the IMD layer **608** and extend lengthwise in the Y-direction. The metal conductors **1306** overlap cell short boundaries of the cell boundary CB of the SRAM cell **101**I in the Y-direction, as shown in FIG. **16**B. In some embodiments, the metal conductors **1306-1** and **1306-2** serve as the VSS lines that are electrically coupled to a voltage node or voltage source (not shown) (e.g., the voltage node (or voltage source) VSS discussed above) and electrically connected to the source/drain features of the pull-down transistors in the same column of the array of the SRAM cell. As shown in FIGS. **16**A and **16**B, in the SRAM cell **101**I, the metal conductor **1306-1** and **1306-2** are respectively under the pull-down transistor PD-**1** and PD-**2**. The metal conductor **1306-1** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**1** through the via **1304-1** and the source/drain contact **1302-1**; and the metal conductor **1306-2** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**2** through the via **1304-2** and the source/drain contact **1302-2**. In some embodiments, the metal conductors **1306-1** and **1306-2** may be referred to as (back-side) VSS conductors or (back-side) VSS lines.
(115) FIGS. **17**A and **17**B illustrate top views (or layouts) of an SRAM cell **101**J that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some

embodiments of the present disclosure. FIG. **17**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **17**B illustrates the features in the device region and the back-side interconnection structure.

(116) The SRAM cell **101**J is similar to the SRAM cell **101**I discussed above, except that the SRAM cell **101**J further includes vias **1402** (including vias **1402-1** and **1402-2**) and a metal conductor **1404**, which are under (or at the back-side of) the transistors in the SRAM cell **101**J (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).

(117) The metal conductor **1404** is in a metal layer BML**2** in the IMD layer **608** and extend lengthwise in the X-direction. The metal layer BML**2** is under the metal layer BML**1**, and thus the metal conductor **1404** is under the metal conductors **606** and **1306**. The metal conductor **1404** is electrically connected to the metal conductors **1306-1** and **1306-2** to serve as the power mesh line for connecting the voltage node VSS to the metal conductors **1306-1** and **1306-3**. As shown in FIGS. **17**A and **17**B, in the SRAM cell **101**J, the metal conductor **1404** is electrically connected to the metal conductor **1306-1** through the via **1402-1** and electrically connected to the metal conductor **1306-2** through the via **1402-2**. In some embodiments, the metal conductor **1404** may be referred to as (front-side) VSS power mesh line or VSS power mesh conductor.

(118) FIGS. **18**A and **18**B illustrate top views (or layouts) of an SRAM cell **101**K that can be one embodiment of the SRAM cells **101** implemented in the array **100**, in accordance with some embodiments of the present disclosure. FIG. **18**A illustrates the features in the device region (including transistors) and the front-side interconnection structure (including vias and metal conductors), and FIG. **18**B illustrates the features in the device region and the back-side interconnection structure.

(119) The SRAM cell **101**K is similar to the SRAM cell **101**A discussed above, except that the metal conductor for word-line is formed in the metal layer BML**1**, the metal conductors for VSS lines are formed in the metal layer BML**2**, and no metal conductor for VDD line in the front-side interconnection structure **502**. The vias **504**, the metal conductors **506**, the vias **508**, and the metal conductors **510** shown in the SRAM cell **101**A are not formed in the SRAM cell **101**G. The SRAM cell **101**K further includes a metal conductor **1502**, vias **1504** (including vias **1504-1** and **1504-2**), metal conductors **1506** (including metal conductors **1506-1** and **1506-2**) vias **1508** (including vias **1508-1** and **1508-2**), and metal conductors **1510** (including metal conductors **1510-1** and **1510-2**) in the front-side interconnection structure **502**, which are over (or at the back-side of) the transistors in the SRAM cell **101**K (e.g., the pass-gate transistors PG-**1** and PG-**2**, the pull-down transistors PD-**1** and PD-**2**, and the pull-up transistors PU-**1** and PU-**2**).

(120) The metal conductor **1502** is in the metal layer ML**1** in the IMD layer **516** and extend lengthwise in the X-direction. In some embodiments, the metal conductor **1502** serves as the word-line that is electrically connected to a controller (e.g., the controller **103** discussed above) and electrically connected to the gate structures (more specifically, the gate electrodes) of the pass-gate transistors in the same row of the array of the SRAM cell. As shown in FIGS. **18**A to **18**B, in the SRAM cell **101**K, the metal conductor **1502** is over the pass-gate transistor PG-**1** and PG-**2**. The metal conductor **1502** is electrically connected to the gate structure **404-1** of the pass-gate transistor PG-**1** through the gate via **512-1**, and is electrically connected to the gate structure **404-4** of the pass-gate transistor PG-**2** through the gate via **512-2**. In some embodiments, the metal conductor **1502** may be referred to as (front-side) word-line conductor.

(121) The metal conductors **1506** are in the metal layer ML**1** in the IMD layer **516** and extend lengthwise in the X-direction. The metal conductors **1510** are in the metal layer ML**2** in the IMD layer **516** and extend lengthwise in the Y-direction. The metal conductors **1510** overlap cell short boundaries of the cell boundary CB of the SRAM cell **101**K in the Y-direction, as shown in FIG. **18**A. In some embodiments, the metal conductors **1510-1** and **1510-2** serve as the VSS lines that

are electrically coupled to a voltage node or voltage source (not shown) (e.g., the voltage node (or voltage source) VSS discussed above) and electrically connected to the source/drain features of the pull-down transistors in the same column of the array of the SRAM cell. As shown in FIGS. **18**A and **18**B, in the SRAM cell **101**K, the metal conductor **1510-1** and **1510-2** are respectively over the pull-down transistor PD-**1** and PD-**2**. The metal conductor **1510-1** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**1** through the via **1508-1**, the metal conductor **1506-1**, the via **1504-1**, and the source/drain contact **430-5**; and the metal conductor **1510-2** is electrically connected to the source/drain features **412**N of the pull-down transistor PD-**2** through the via **1508-2**, the metal conductor **1506-2**, the via **1504-2**, and the source/drain contact **430-2**. In some embodiments, the metal conductors **1510-1** and **1510-2** may be referred to as (front-side) VSS conductors or (front-side) VSS lines.

(122) As shown in FIG. **18**A, the metal conductor **1502** for word-line is non-rectangular shape. The metal conductor **1502** further includes protrusion portions **1602-1** and **1602-2**. The protrusion portions **1602-1** and **1602-2** extend lengthwise in the X-direction and protrude from the metal conductor **1502** in the Y-direction. In some embodiments, the protrusion portions **1602-1** and **1602-2** are respectively separated from the active areas **402-3** and **402-2** in the top view. Such metal conductor **1502** with protrusion portions **1602-1** and **1602-2** may have lower resistance because the area of the metal conductor **1502** is enlarged.

(123) The embodiments disclosed herein relate to memory structures, and more particularly to memory structures comprising the metal conductors for the bit-line and bit-line-bar that are under or at the back-side of the SRAM cells (more specifically, functional transistors). Furthermore, the present embodiments provide one or more of the following advantages. The metal conductors for bit-line and bit-line-bar at the back-side provides a reduced routing complexity for the SRAM cells, a lower circuit resistance, and a lower parasitic capacitance, which improves the performance of the SRAM cells, such as RC delay. Furthermore, the gate structures do not affect (back-side) source/drain contacts electrically connected to the metal conductors for bit-line and bit-line-bar, thereby improving the reliability of the read/write operations of the SRAM cells.

(124) Thus, one of the embodiments of the present disclosure describes a memory structure that includes a static random-access memory (SRAM) cell, a word-line conductor extending in a first direction, a first source/drain contact, a second source/drain contact, a bit-line conductor extending in a second direction, and a bit-line-bar conductor extending in the second direction. The second direction is perpendicular to the first direction. The SRAM cell has a first pass-gate transistor and a second pass-gate transistor. The word-line conductor is over and electrically connected to gate electrodes of the first pass-gate transistor and the second pass-gate transistor. The first source/drain contact is under and electrically connected to a source/drain feature of the first pass-gate transistor. The second source/drain contact is under and electrically connected to a source/drain feature of the second pass-gate transistor. The bit-line conductor is under and electrically connected to the first source/drain contact. The bit-line conductor is under and electrically connected to the second source/drain contact.

(125) In some embodiments, the bit-line conductor is in contact with the first source/drain contact and the bit-line-bar conductor is in contact with the second source/drain contact.

(126) In some embodiments, the memory structure further includes a first via vertically between the first source/drain contact and the bit-line conductor, and a second via vertically between the second source/drain contact and the bit-line-bar conductor. The bit-line conductor is electrically connected to the first source/drain contact through the first via. The bit-line conductor is electrically connected to the second source/drain contact through the first via.

(127) In some embodiments, the SRAM cell further includes a first pull-up transistor and a second pull-up transistor. The memory structure further includes a first VDD conductor extending in the second direction and under the first pull-up transistor and the second pull-up transistor. The first VDD conductor is electrically connected source/drain features of the first pull-up transistor and the

second pull-up transistor.
(128) In some embodiments, the memory structure further includes a second VDD conductor extending in the second direction and over the first pull-up transistor and the second pull-up transistor. The first VDD conductor is electrically connected to the source/drain features of the first pull-up transistor and the second pull-up transistor.
(129) In some embodiments, the SRAM cell further includes a first pull-down transistor and a second pull-down transistor. The memory structure further includes a first VSS conductor extending in the second direction and electrically connected to a source/drain feature of the first pull-down transistor, and a second VSS conductor extending in the second direction and electrically connected to a source/drain feature of the second pull-down transistor.
(130) In some embodiments, the first VSS conductor and the second VSS conductor are respectively under the first pull-down transistor and the second pull-down transistor.
(131) In some embodiments, the first VSS conductor and the second VSS conductor are respectively over the first pull-down transistor and the second pull-down transistor.
(132) In some embodiments, the memory structure further includes a VSS power mesh conductor extending in the first direction. The VSS power mesh conductor is over and electrically connected to the first VSS conductor and the second VSS conductor.
(133) In some embodiments, the memory structure further includes a first word-line landing pad extending in the second direction and over the first pass-gate transistor, and a second word-line landing pad extending in the second direction and over the second pass-gate transistor. The word-line conductor is electrically connected to the gate electrodes of the first pass-gate transistor and the second pass-gate transistor respectively through the first word-line landing pad and the second word-line landing pad.
(134) In another of the embodiments, discussed is a memory structure including a static random-access memory (SRAM) cell, a first metal layer under the SRAM cell, a second metal layer over the SRAM cell, and a third metal layer over the second metal layer. The SRAM cell has a first pass-gate transistor, a second pass-gate transistor, a first pull-down transistor, and a second pull-down transistor. The first metal layer includes a bit-line conductor and a bit-line-bar conductor extending in a first direction. The bit-line conductor is electrically connected to a source/drain feature of the first pass-gate transistor. The bit-line-bar conductor is electrically connected to a source/drain feature of the second pass-gate transistor. The second metal layer includes a first VSS conductor and a second VSS conductor extending in the first direction. The first VSS conductor is electrically connected to a source/drain feature of the first pull-down transistor. The second VSS conductor is electrically connected to a source/drain feature of the second pull-down transistor. The third metal layer includes a word-line conductor extending in a second direction. The second direction is perpendicular to the first direction. The word-line conductor is electrically connected to gate electrodes of the first pass-gate transistor and the second pass-gate transistor.
(135) In some embodiments, the SRAM cell further includes a first pull-up transistor and a second pull-up transistor. The first metal layer further includes a first VDD conductor extending in the first direction and electrically connected to source/drain features of the first pull-up transistor and the second pull-up transistor.
(136) In some embodiments, the second metal layer further includes a second VDD conductor extending in the first direction and electrically connected to the source/drain features of the first pull-up transistor and the second pull-up transistor.
(137) In some embodiments, the first metal layer further includes a third VSS conductor and a fourth VSS conductor extending in the first direction. The third VSS conductor is electrically connected to the source/drain feature of the first pull-down transistor. The fourth VSS conductor is electrically connected to the source/drain feature of the second pull-down transistor.
(138) In some embodiments, the memory structure further includes a fourth metal layer under the first metal layer. The fourth metal layer includes a VSS power mesh conductor extending in the

second direction. The VSS power mesh conductor is electrically connected to the third VSS conductor and the fourth VSS conductor.

(139) In some embodiments, the third VSS conductor and the fourth VSS conductor overlap cell short boundaries of the SRAM cell.

(140) In yet another of the embodiments, discussed is a memory structure that includes an array of static random-access memory (SRAM) cells, bit-line conductors and bit-line-bar conductors, first VSS conductors and second VSS conductors, word-line conductors, first VDD conductors, and second VDD conductors. Each of the SRAM cells comprises a first pass-gate transistor, a second pass-gate transistor, a first pull-down transistor, a second pull-down transistor, a first pull-up transistor, and a second pull-up transistor. The bit-line conductors and the bit-line-bar conductors in a first metal layer under the SRAM cell and extend in a first direction. The bit-line conductors are electrically connected to source/drain features of the first pass-gate transistors. The bit-line-bar conductors are electrically connected to source/drain features of the second pass-gate transistors. The first VSS conductors and the second VSS conductors in a second metal layer over the SRAM cell and extend in the first direction. The first VSS conductors are electrically connected to source/drain features of the first pull-down transistors. The second VSS conductors are electrically connected to source/drain features of the second pull-down transistors. The word-line conductors in a third metal layer over the second metal layer and extend in a second direction perpendicular to the first direction. The word-line conductors are electrically connected to gate electrodes of the first pass-gate transistors and the second pass-gate transistors. The first VDD conductors in the second metal layer and extend in the first direction. The first VDD conductors are electrically connected to source/drain features of the first pull-up transistor and the second pull-up transistor. The second VDD conductors in the first metal layer and extend in the first direction. The second VDD conductors are electrically connected to the source/drain features of the first pull-up transistor and the second pull-up transistor.

(141) In some embodiments, the memory structure further includes an edge strap region on an edge of the array of the SRAM cells. The edge strap region includes front-side bit-line conductors in the second metal layer, and bit-line tap structures electrically connecting bit-line conductors to the front-side bit-line conductors.

(142) In some embodiments, the memory structure further includes VSS power mesh conductors in the third metal layer and extending in the second direction. The VSS power mesh conductors are electrically connected to the first VSS conductors and the second VSS conductors.

(143) In some embodiments, the word-line conductors in the third metal layer are first word-line conductors. The memory structure further includes second word-line conductors extending in the second direction and over the first word-line conductors. The second word-line conductors are electrically connected to the first word-line conductors.

(144) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A memory structure, comprising: a static random-access memory (SRAM) cell having a first pass-gate transistor and a second pass-gate transistor; a word-line conductor extending in a first direction, wherein the word-line conductor is over and electrically connected to gate electrodes of

the first pass-gate transistor and the second pass-gate transistor; a first source/drain contact under and electrically connected to a source/drain feature of the first pass-gate transistor; a second source/drain contact under and electrically connected to a source/drain feature of the second pass-gate transistor; a bit-line conductor extending in a second direction perpendicular to the first direction, wherein the bit-line conductor is under and electrically connected to the first source/drain contact; a bit-line-bar conductor extending in the second direction, wherein the bit-line-bar conductor is under and electrically connected to the second source/drain contact; a first word-line landing pad extending in the second direction and over the first pass-gate transistor; and a second word-line landing pad extending in the second direction and over the second pass-gate transistor, wherein the word-line conductor is electrically connected to the gate electrodes of the first pass-gate transistor and the second pass-gate transistor respectively through the first word-line landing pad and the second word-line landing pad.

2. The memory structure of claim 1, wherein the bit-line conductor is in contact with the first source/drain contact and the bit-line-bar conductor is in contact with the second source/drain contact.

3. The memory structure of claim 1, further comprising: a first via vertically between the first source/drain contact and the bit-line conductor, wherein the bit-line conductor is electrically connected to the first source/drain contact through the first via; and a second via vertically between the second source/drain contact and the bit-line-bar conductor, wherein the bit-line-bar conductor is electrically connected to the second source/drain contact through the second-first via.

4. The memory structure of claim 1, wherein the SRAM cell further comprises a first pull-up transistor and a second pull-up transistor, and wherein the memory structure further comprises: a first VDD conductor extending in the second direction and under the first pull-up transistor and the second pull-up transistor, wherein the first VDD conductor is electrically connected source/drain features of the first pull-up transistor and the second pull-up transistor.

5. The memory structure of claim 4, further comprising: a second VDD conductor extending in the second direction and over the first pull-up transistor and the second pull-up transistor, wherein the second VDD conductor is electrically connected to the source/drain features of the first pull-up transistor and the second pull-up transistor.

6. The memory structure of claim 1, wherein the SRAM cell further comprises a first pull-down transistor and a second pull-down transistor, and wherein the memory structure further comprises: a first VSS conductor extending in the second direction and electrically connected to a source/drain feature of the first pull-down transistor; and a second VSS conductor extending in the second direction and electrically connected to a source/drain feature of the second pull-down transistor.

7. The memory structure of claim 6, wherein the first VSS conductor and the second VSS conductor are respectively under the first pull-down transistor and the second pull-down transistor.

8. The memory structure of claim 6, wherein the first VSS conductor and the second VSS conductor are respectively over the first pull-down transistor and the second pull-down transistor.

9. The memory structure of claim 8, further comprising: a VSS power mesh conductor extending in the first direction, wherein the VSS power mesh conductor is over and electrically connected to the first VSS conductor and the second VSS conductor.

10. A memory structure, comprising: a static random-access memory (SRAM) cell having a first pass-gate transistor, a second pass-gate transistor, a first pull-down transistor, and a second pull-down transistor; a first metal layer under the SRAM cell, wherein the first metal layer comprises a bit-line conductor and a bit-line-bar conductor extending in a first direction, wherein the bit-line conductor is electrically connected to a source/drain feature of the first pass-gate transistor and the bit-line-bar conductor is electrically connected to a source/drain feature of the second pass-gate transistor; a second metal layer over the SRAM cell, wherein the second metal layer comprises a first VSS conductor and a second VSS conductor extending in the first direction, wherein the first VSS conductor is electrically connected to a source/drain feature of the first pull-down transistor

and the second VSS conductor is electrically connected to a source/drain feature of the second pull-down transistor; and a third metal layer over the second metal layer, wherein the third metal layer comprises a word-line conductor extending in a second direction perpendicular to the first direction, wherein the word-line conductor is electrically connected to gate electrodes of the first pass-gate transistor and the second pass-gate transistor.

11. The memory structure of claim 10, wherein the SRAM cell further comprises a first pull-up transistor and a second pull-up transistor, and wherein the first metal layer further comprises a first VDD conductor extending in the first direction and electrically connected to source/drain features of the first pull-up transistor and the second pull-up transistor.

12. The memory structure of claim 11, wherein the second metal layer further comprises a second VDD conductor extending in the first direction and electrically connected to the source/drain features of the first pull-up transistor and the second pull-up transistor.

13. The memory structure of claim 10, wherein the first metal layer further comprises a third VSS conductor and a fourth VSS conductor extending in the first direction, wherein the third VSS conductor is electrically connected to the source/drain feature of the first pull-down transistor and the fourth VSS conductor is electrically connected to the source/drain feature of the second pull-down transistor.

14. The memory structure of claim 13, further comprising: a fourth metal layer under the first metal layer, wherein the fourth metal layer comprises a VSS power mesh conductor extending in the second direction, wherein the VSS power mesh conductor is electrically connected to the third VSS conductor and the fourth VSS conductor.

15. The memory structure of claim 13, wherein the third VSS conductor and the fourth VSS conductor overlap cell short boundaries of the SRAM cell.

16. The memory structure of claim 10, further comprising: a first word-line landing pad extending in the first direction and over the first pass- gate transistor; and a second word-line landing pad extending in the first direction and over the second pass-gate transistor, wherein the word-line conductor is electrically connected to the gate electrodes of the first pass-gate transistor and the second pass-gate transistor respectively through the first word-line landing pad and the second word-line landing pad.

17. A memory structure, comprising: an array of static random-access memory (SRAM) cells, wherein each of the SRAM cells comprises a first pass-gate transistor, a second pass-gate transistor, a first pull-down transistor, a second pull-down transistor, a first pull-up transistor, and a second pull-up transistor; bit-line conductors and bit-line-bar conductors in a first metal layer under the SRAM cells and extending in a first direction, wherein the bit-line conductors are electrically connected to source/drain features of the first pass-gate transistors and the bit-line-bar conductors are electrically connected to source/drain features of the second pass-gate transistors; first VSS conductors and second VSS conductors in a second metal layer over the SRAM cells and extending in the first direction, wherein the first VSS conductors are electrically connected to source/drain features of the first pull-down transistors and the second VSS conductors are electrically connected to source/drain features of the second pull-down transistors; word-line conductors in a third metal layer over the second metal layer and extending in a second direction perpendicular to the first direction, wherein the word-line conductors are electrically connected to gate electrodes of the first pass-gate transistors and the second pass-gate transistors; first VDD conductors in the second metal layer and extending in the first direction, wherein the first VDD conductors are electrically connected to source/drain features of the first pull-up transistor and the second pull-up transistor; and second VDD conductors in the first metal layer and extending in the first direction, wherein the second VDD conductors are electrically connected to the source/drain features of the first pull-up transistor and the second pull-up transistor.

18. The memory structure of claim 17, further comprising: an edge strap region on an edge of the array of the SRAM cells, wherein the edge strap region comprises: front-side bit-line conductors in

the second metal layer; and bit-line tap structures electrically connecting the bit-line conductors to the front-side bit-line conductors.

19. The memory structure of claim 17, further comprising: VSS power mesh conductors in the third metal layer and extending in the second direction, wherein the VSS power mesh conductors are electrically connected to the first VSS conductors and the second VSS conductors.

20. The memory structure of claim 17, wherein the word-line conductors in the third metal layer are first word-line conductors, wherein the memory structure further comprises: second word-line conductors extending in the second direction and over the first word-line conductors, wherein the second word-line conductors are electrically connected to the first word-line conductors.